

Supplemental Tables

For: Hyun-Jung Shin, Sun Woo Nam, Heeyeon Kim et al. Postoperative delirium following dexmedetomidine versus propofol sedation in healthy older adults undergoing orthopedic lower limb surgery with spinal anesthesia: A randomized controlled trial

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Table S1. Characteristics of patients

	Intention-to-Treat		Per-Protocol	
	Propofol (n = 366)	Dexmedetomidine (n = 366)	Propofol (n = 341)	Dexmedetomidine (n = 342)
Age (year)	71 (67–75)	72 (68–76)	71 (67–76)	72 (68–76)
Sex				
Male	77 (21%)	78 (21%)	77 (23%)	70 (21%)
Female	289 (79%)	288 (79%)	264 (77%)	272 (79%)
Weight (kg)	63 (56–69)	61 (56–69)	63 (56–69)	61 (56–69)
Height (cm)	154 (150–159)	153 (149–159)	154 (150–159)	153 (149–159)
ASA classification				
1	66 (18%)	52 (14%)	60 (18%)	47 (14%)
2	300 (82%)	314 (86%)	281 (82%)	295 (86%)
Underlying disease				
Hypertension	223 (61%)	232 (64%)	205 (60%)	215 (63%)
Diabetes mellitus	77 (21%)	101 (28%)	70 (21%)	92 (27%)
Chronic kidney disease	7 (1.9%)	8 (2.2%)	5 (1.5%)	6 (1.8%)
Chronic liver disease	3 (0.8%)	4 (1.1%)	3 (0.9%)	2 (0.6%)
Chronic pulmonary disease	8 (2.2%)	2 (0.5%)	8 (2.3%)	2 (0.6%)
Rheumatoid arthritis	4 (1.1%)	13 (3.6%)	4 (1.2%)	10 (2.9%)

Data expressed as median (interquartile range), or n (%).

ASA, American Society of Anesthesiologists

Table S2. Perioperative surgical and anesthetic variables

	Intention-to-Treat			Per-Protocol		
	Propofol (n = 366)	Dexmedetomidine (n = 366)	P Value	Propofol (n = 341)	Dexmedetomidine (n = 342)	P Value
Type of surgery						
Hip and femur	92 (25%)	89 (24%)	0.667	85 (25%)	76 (22%)	0.418
Knee and tibia/fibula	248 (68%)	247 (68%)	0.937	231 (68%)	238 (70%)	0.621
Ankle and foot	26 (7%)	30 (8%)	0.472	25 (7%)	28 (8%)	0.775
Surgery time (min)	85 (75–105)	85 (75–105)	0.244	85 (75–105)	85 (75–105)	0.402
Anesthesia time (min)	125 (115–150)	130 (115–155)	0.413	125 (115–150)	130 (115–155)	0.668
Estimated blood loss (mL)	50 (50–250)	50 (50–200)	0.336	50 (50–250)	50 (50–200)	0.222
Crystalloid (mL)	450 (300–600)	400 (300–550)	0.258	450 (300–600)	400 (300–550)	0.173
Colloid (mL)	0 (0–0)	0 (0–100)	0.651	0 (0–50)	0 (0–0)	0.959
Urine (mL)	244 (130–400)	250 (150–380)	0.778	255 (150–414)	259 (169–398)	0.785
Red blood cell (unit)						
Before surgery	0 (0–0)	0 (0–0)	>0.999	0 (0–0)	0 (0–0)	>0.999
During surgery	0 (0–0)	0 (0–0)	0.799	0 (0–0)	0 (0–0)	0.804
After surgery	0 (0–0)	0 (0–0)	0.635	0 (0–0)	0 (0–0)	0.640
Modified observer's assessment of alertness/sedation score at PACU	5 (5–5)	5 (4–5)	<0.001	5 (5–5)	5 (4–5)	<0.001
PACU stay (min)	27 (20–43)	37 (23–60)	<0.001	26 (20–43)	37 (23–60)	<0.001
Length of stay (day)						
Before surgery	1 (1–1)	1 (1–1)	0.870	1 (1–1)	1 (1–1)	0.782
After surgery	6 (6–7)	6 (6–7)	0.778	6 (6–7)	6 (6–7)	0.537
Total	7 (7–8)	6 (6–7)	0.821	7 (7–8)	7 (7–8)	0.626
ICU care, postoperatively	1 (0.3%)	0 (0%)	>0.999	1 (0.3%)	0 (0%)	0.499
Ephedrine, total (mg)	5 (0–5)	5 (0–10)	0.179	5 (0–5)	5 (0–10)	0.181
Phenylephrine, total (µg)	0 (0–30)	0 (0–20)	0.411	0 (0–50)	0 (0–30)	0.223
Atropine, total (mg)	0 (0–0)	0 (0–0)	0.314	0 (0–0)	0 (0–0)	0.207
MAP (mmHg)						
Before sedation	94 (87–102 [61–122])	92 (83–101 [61–121])	0.098	94 (87–103 [61–125])	92 (84–101 [61–124])	0.083

During sedation	74 (69–79 [41–119])	77 (71–84 [53–103])	<0.001	74 (69–79 [41–109])	77 (70–83 [53–103])	<0.001
At PACU	80 (74–87 [60–112])	74 (68–80 [55–112])	<0.001	79 (74–87 [60–112])	73 (70–80 [55–107])	<0.001
HR (rates/min)						
Before sedation	74 (67–82 [46–112])	74 (65–82 [45–125])	0.194	74 (67–82 [46–112])	73 (62–83 [45–125])	0.130
During sedation	63 (58–70 [45–125])	60 (55–66 [44–104])	<0.001	63 (58–70 [45–100])	60 (54–65 [44–104])	<0.001
At PACU	68 (62–77 [44–100])	64 (58–72 [43–99])	<0.001	68 (62–77 [44–99])	64 (58–71 [43–99])	<0.001

Data expressed as median (interquartile range), median (interquartile range [range]), or n (%).

MAP; mean arterial pressure; HR, heart rates; PACU, postanesthesia care unit; ICU, intensive care unit

Table S3. Basal cognitive status and postoperative delirium

	Intention-to-Treat				Per-Protocol			
	Propofol (n = 366)	Dexmedetomidine (n = 366)	Odds Ratio (95% CI)	P-value	Propofol (n = 341)	Dexmedetomidine (n = 342)	Odds Ratio (95% CI)	P-value
Mini-Cog score	5 (3–5)	5 (3–5)		0.102	4 (3–5)	4 (3–5)		0.118
Delirium*	24 (6.6%)	11 (3.0%)	0.42 (0.201–0.86)	0.036	23 (6.7%)	10 (2.9%)	0.368 (0.175–0.77)	0.021
Delirium onset				0.802				0.606
Postoperative 1 day	15 (4.1%)	7 (1.9%)			14 (4.1%)	6 (1.8%)		
Postoperative 2 day	5 (1.4%)	3 (0.8%)			5 (1.5%)	3 (0.9%)		
Postoperative 3 day	4 (1.1%)	1 (0.3%)			4 (1.2%)	1 (0.3%)		
Delirium period**				Not applicable				Not applicable
Postoperative 1 to 2 days	6 (1.6%)	0 (0%)			5 (1.5%)	0 (0%)		
Postoperative 2 to 3 days	1 (0.3%)	1 (0.3%)			0 (0%)	1 (0.3%)		
Postoperative 1 to 3 days	1 (0.3%)	0 (0%)			0 (0%)	0 (0%)		

Data expressed as median (interquartile range), n (%), or odd ratio (95% CI).

CI, confidence interval.

* For patients with missing data due to early hospital discharge, if they were evaluated as positive for the confusion assessment method even once during the postoperative admission period, they were classified as postoperative delirium: In the propofol group, 8 and 16 patients discharged at 2 and 3 days after surgery, respectively. In the dexmedetomidine group, 4 and 18 patients discharged at 2 and 3 days after surgery, respectively.

Per-Protocol population: In the propofol group, 8 and 16 patients discharged at 2 and 3 days after surgery, respectively. In the dexmedetomidine group, 3 and 17 patients discharged at 2 and 3 days after surgery, respectively.

** Only patients with delirium lasting more than two days were included.